CSE381

INTRODUCTION TO MACHINE LEARNING

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MILESTONE 1

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1. Problem Framing and Data Understanding

1.1 Project Objective

The objective of this project is to develop a supervised machine learning solution that predicts whether a mushroom is edible or poisonous based on its physical characteristics. Correctly identifying poisonous mushrooms is important because an incorrect prediction may have serious consequences.

The input variables describe characteristics such as cap shape, cap color, gill size, stalk properties, ring type, population, and habitat. The target variable represents mushroom edibility and contains two possible classes:

- e: Edible
- p: Poisonous

Since the target variable contains two classes, this task is a supervised binary classification problem.

1.2 Anticipated Dataset Challenges

Before preprocessing, the following challenges were anticipated:

- All input features are categorical and must be converted into numerical form.
- Some features may contain missing values.
- Certain features may be strongly related to the target and make the classification problem trivial.
- The target classes may be imbalanced.
- Separate preprocessing of the training and testing data must be handled carefully to prevent data leakage.

These challenges were investigated during the initial data inspection and addressed in the preprocessing stage.

1.3 Code Google Colab

https://colab.research.google.com/drive/1gHk7dHS_0-hDRb7unL6lWuB5pIbFG4ar?usp=sharing

2. Data Acquisition and Documentation

2.1 Dataset Source

The Mushroom Edibility Dataset was obtained from the UCI Machine Learning Repository:

https://archive.ics.uci.edu/dataset/73/mushroom

The dataset was loaded directly into the notebook using the ucimlrepo Python package. No external or manually modified version of the dataset was used.

2.2 Dataset Profile

Property	**Description**
Dataset name	Mushroom Edibility Dataset
Source	UCI Machine Learning Repository
Number of samples	8,124
Number of original input features	22
Feature types	Categorical
Target variable	Mushroom edibility
Target classes	Edible and Poisonous
Classification type	Binary classification

The input features were stored in X, while the target variable was stored separately in y. This separation made the preprocessing and train-test splitting stages easier to manage.

3. Data Cleaning and Preprocessing

3.1 Initial Data Inspection

The dataset was inspected using several Pandas methods. The head() method was used to preview the first observations, while info() was used to examine the number of entries, feature names, non-null counts, and data types.

The initial inspection confirmed that the dataset contains 8,124 samples and 22 categorical input features. All features had the object data type because their values were represented using categorical codes.

Missing values were checked using isnull().sum(), duplicates were checked using duplicated().sum(), and the target distribution was examined using value_counts().

3.2 Duplicate Records

The duplicate check showed that the original dataset contained zero duplicated rows. Therefore, no observations were removed during this step.

Keeping all non-duplicate observations preserves the full amount of available training information.

3.3 Missing Values

The missing-value inspection showed that stalk-root was the only incomplete feature. It contained 2,480 missing values, while the remaining features contained no missing values.

Data-quality check	Result
Total samples	8,124
Duplicate rows	0
Features containing missing values	1
Missing values in stalk-root	2,480

Deleting every row with a missing stalk-root value would remove a considerable portion of the dataset. Therefore, imputation was selected instead of row deletion.

3.4 Target Distribution and Class Balance

The distribution of the target variable was examined before model training.

Class	Number of samples	Percentage
Edible (e)	4,208	51.8%
Poisonous (p)	3,916	48.2%

The difference between the two classes is small, indicating that the dataset is approximately balanced. As a result, no oversampling, undersampling, or synthetic balancing technique was required.

A stratified train-test split was still used to preserve this distribution in both subsets.

3.5 Feature Reduction

The project requirements specify that a few “too obvious” features must be removed so that the classification task does not become trivial. Accordingly, the following two features were excluded:

- odor
- spore-print-color

odor is closely associated with mushroom edibility and can act as a very strong direct indicator of the target. Similarly, spore-print-color contains highly distinctive category information that may dominate the classification decision.

Removing these features makes the experiment more meaningful because the models must learn from combinations of the remaining mushroom characteristics rather than depending mainly on a small number of obvious predictors.

After feature reduction, the dataset contained 20 original categorical input features.

3.6 Train-Test Splitting Strategy

The data was divided into training and testing sets using an 80/20 split.

```
train_test_split(
  X,
  y,
  test_size=0.2,
  random_state=42,
  stratify=y
)
```

The resulting split was:

Dataset portion	Number of samples	Percentage
Training set	6,499	80%
Testing set	1,625	20%

The training set is used to learn preprocessing decisions and train the classifier. The testing set remains separate and is used only for the final evaluation of the trained model.

A fixed random_state of 42 was used to make the split reproducible. The stratify=y parameter maintained approximately the same edible-poisonous distribution in both subsets.

3.7 Missing-Value Imputation

The missing values in stalk-root were handled using mode imputation. This approach replaces missing entries with the most frequently occurring category.

The replacement value was calculated using only X_train. The same training-derived value was then applied to both X_train and X_test.

This strategy is important because using information from the testing set to calculate the replacement value could introduce data leakage. After imputation, both datasets contained zero missing values.

3.8 Categorical Feature Encoding

Machine learning classifiers generally require numerical input. Therefore, the remaining categorical features were converted using One-Hot Encoding through the Pandas get_dummies() method.

One-Hot Encoding creates a separate binary column for each category. In the resulting data:

- True indicates that the category is present.
- False indicates that the category is absent.

The training and testing sets were encoded separately. Their columns were then aligned to guarantee that both sets contained the same encoded features in the same order. Any encoded column that was not present in one set was assigned a value of zero.

3.9 Final Preprocessed Data

After missing-value handling and One-Hot Encoding, the final dataset dimensions were:

Dataset	Samples	Encoded features	Missing values
Training data	6,499	98	0
Testing data	1,625	98	0

The preprocessing stage produced clean, consistent, and model-ready data. The final training and testing sets contain matching feature structures, no missing values, and approximately balanced target classes.

4. Exploratory Data Analysis

Exploratory Data Analysis was conducted to understand the target distribution and investigate meaningful relationships between mushroom characteristics and edibility before training the baseline classifier.

Since all original input features are categorical, count-based bar charts were used to compare the frequencies of edible and poisonous mushrooms. Four visualizations were created, satisfying the requirement of exactly three to five meaningful and captioned figures.

The analysis examined the target distribution and three selected features: gill-size, bruises, and habitat. The removed features, odor and spore-print-color, were not included because they had already been excluded during the feature-reduction stage.

Consistent colors were used throughout the figures: green represents edible mushrooms, while red represents poisonous mushrooms. The original category codes were also replaced with descriptive labels to improve figure readability.

4.1 Class Distribution

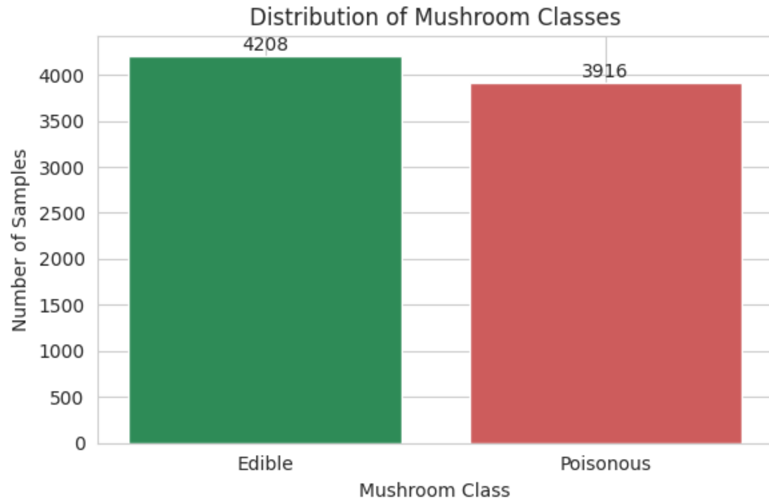

Figure 1: Distribution of edible and poisonous mushrooms.

Figure 1 presents the distribution of the target classes in the complete dataset.

Mushroom class	Number of samples	Percentage
Edible	4,208	51.8%
Poisonous	3,916	48.2%
Total	8,124	100%

The dataset contains 4,208 edible mushrooms and 3,916 poisonous mushrooms. The difference between the two classes is only 292 samples, indicating that the dataset is approximately balanced.

This balance is important because a large class imbalance could cause the model to favor the majority class. In this case, a classifier that always predicts the majority class would achieve only approximately 51.8% accuracy. Therefore, the baseline model must learn meaningful relationships between the features and the target to obtain good performance.

Because no substantial class imbalance was detected, oversampling, undersampling, and synthetic data-generation techniques were not required. However, stratified sampling was used during the train-test split to preserve approximately the same class distribution in both sets.

4.2 First Feature–Target Relationship

Gill Size and Mushroom Edibility

Figure 2: Relationship between gill size and mushroom edibility.

Figure 2 compares the number of edible and poisonous mushrooms across the two gill-size categories: broad and narrow.

Gill size	Edible	Poisonous	Total
Broad	3,920	1,692	5,612
Narrow	288	2,224	2,512

Broad gills are the most common category, appearing in 5,612 mushroom samples. Among broad-gill mushrooms, 3,920 are edible and 1,692 are poisonous. This means that broad gills are more frequently associated with edible mushrooms.

Narrow gills show the opposite pattern. Of the 2,512 mushrooms with narrow gills, 2,224 are poisonous and only 288 are edible. Therefore, approximately 88.5% of narrow-gill mushrooms in this dataset are poisonous.

The clear difference between the two distributions indicates that gill-size is a useful predictor of mushroom edibility. Narrow gills provide a particularly strong indication of the poisonous class.

However, gill size does not provide perfect class separation. Some broad-gill mushrooms are poisonous, and a small number of narrow-gill mushrooms are edible. Therefore, this feature should be combined with other characteristics rather than used as an independent classification rule.

4.3 Second Feature–Target Relationship

Bruising and Mushroom Edibility

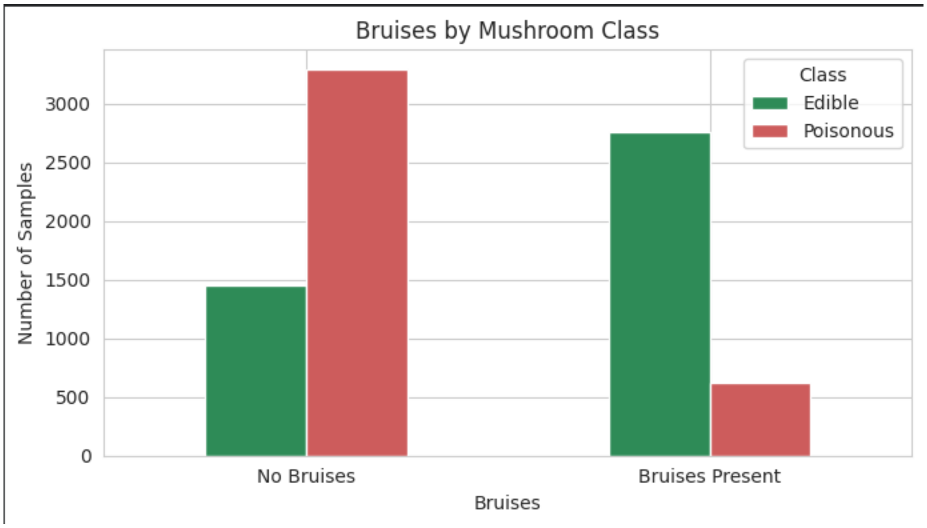

Figure 3: Relationship between bruising and mushroom edibility.

Figure 3 examines the relationship between the presence of bruises and mushroom edibility.

Bruising status	Edible	Poisonous	Total
No bruises	1,456	3,292	4,748
Bruises present	2,752	624	3,376

Among the 4,748 mushrooms without bruises, 3,292 are poisonous and 1,456 are edible. Therefore, approximately 69.3% of mushrooms without bruises are poisonous.

The opposite pattern appears when bruises are present. Out of 3,376 mushrooms with bruises, 2,752 are edible and only 624 are poisonous. This means that approximately 81.5% of mushrooms with bruises are edible.

These results indicate that bruises contains meaningful discriminatory information. The absence of bruises is more strongly associated with poisonous mushrooms, while the presence of bruises is more strongly associated with edible mushrooms.

Nevertheless, bruising status cannot determine edibility independently because both bruising categories contain edible and poisonous samples. This overlap confirms that classification should depend on combinations of several mushroom characteristics.

4.4 Additional Data Relationship

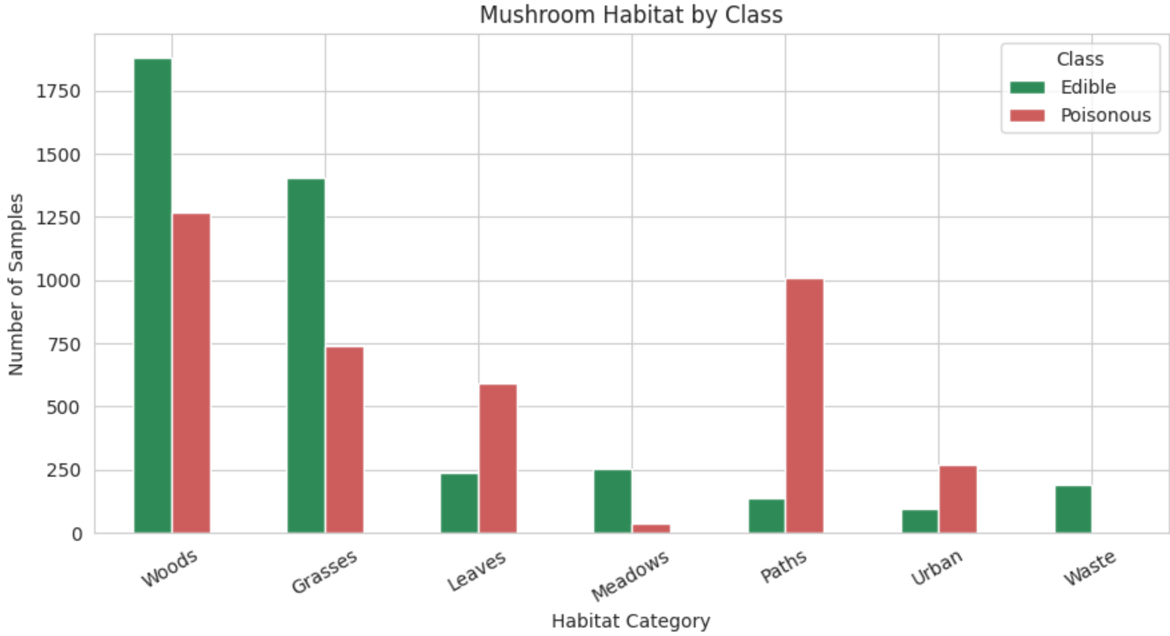

Figure 4: Distribution of mushroom classes across different habitats.

Figure 4 compares mushroom edibility across seven habitat categories: woods, grasses, leaves, meadows, paths, urban environments, and waste areas.

Habitat	Edible	Poisonous	Total
Woods	1,880	1,268	3,148
Grasses	1,408	740	2,148
Leaves	240	592	832
Meadows	256	36	292
Paths	136	1,008	1,144
Urban	96	272	368
Waste	192	0	192

Woods contain the largest number of mushroom samples. Both classes are present in this habitat, although edible mushrooms are more frequent. Grasses show a similar pattern, with 1,408 edible samples and 740 poisonous samples.

Meadows are strongly associated with edible mushrooms in the available dataset. Of the 292 meadow samples, 256 are edible and only 36 are poisonous.

Paths show one of the strongest poisonous patterns. Out of 1,144 mushrooms found near paths, 1,008 are poisonous and only 136 are edible. Therefore, approximately 88.1% of path-habitat mushrooms are poisonous.

Mushrooms found among leaves and in urban environments are also more frequently poisonous. Approximately 71.2% of mushrooms found among leaves and 73.9% of mushrooms found in urban environments are poisonous.

The waste category contains 192 edible samples and no poisonous samples. However, this result should be interpreted cautiously because the waste category contains considerably fewer observations than woods or grasses. The result describes the available dataset and should not be treated as a universal safety rule.

Overall, habitat provides useful contextual information because the class distribution changes considerably across categories. However, several habitats contain both edible and poisonous mushrooms, so habitat alone cannot determine edibility.

The EDA demonstrates that gill-size, bruises, and habitat all have meaningful relationships with the target. Narrow gills, the absence of bruises, and path habitats show strong associations with poisonous mushrooms. In contrast, broad gills, the presence of bruises, and meadow habitats are more commonly associated with edible mushrooms.

None of the selected features perfectly separates the two classes. This supports using a machine learning classifier that combines multiple mushroom characteristics instead of relying on one feature. The results describe associations in the dataset rather than causal relationships or universal biological rules.

5. Baseline Classifier

A single baseline classifier was implemented for the preliminary deliverable. Logistic Regression was selected and trained using the preprocessed data produced in the previous stage.

Only one classifier was included to follow the first-deliverable requirement of implementing exactly one baseline model. Additional classifiers and a complete model comparison will be considered in the final project deliverable.

5.1 Model Selection

Logistic Regression was selected as the baseline classifier because it is a simple, efficient, and widely used method for supervised binary classification.

The model estimates the probability that an observation belongs to a particular class. In this project, it learns the relationship between the encoded mushroom characteristics and the two target classes: edible and poisonous.

Logistic Regression was considered an appropriate baseline for several reasons:

- The problem has two target classes.
- The preprocessed features are numerical binary indicators produced through One-Hot Encoding.
- The model is computationally efficient for the dataset size.
- It provides probability estimates required for ROC and AUC evaluation.
- It offers a clear reference point for comparing more advanced models in the final project.

The model was initialized with max_iter=1000 to provide sufficient iterations for convergence. No hyperparameter tuning was performed because the purpose of the preliminary model was to establish baseline performance.

5.2 Training and Cross-Validation Strategy

The Logistic Regression model was evaluated using five-fold cross-validation on the training set.

In five-fold cross-validation, the training data is divided into five approximately equal subsets. The model is trained five times. During each iteration, four folds are used for training and the remaining fold is used for validation. Each fold is used for validation once.

The cross-validation accuracy scores were:

Fold	Accuracy
Fold 1	99.77%
Fold 2	99.85%
Fold 3	99.38%

Fold 4 99.15%

Fold 5 99.31%

Mean 99.49%

The individual scores range from approximately 99.15% to 99.85%, with a mean cross-validation accuracy of approximately 99.49%.

The consistently high scores across all five folds indicate that the model's performance is stable across different subsets of the training data. The small variation between the folds suggests that the result is not dependent on one favorable validation split.

After cross-validation, the Logistic Regression model was fitted using the complete training set of 6,499 samples. The separate testing set of 1,625 samples was then used for the final baseline evaluation.

6. Performance Evaluation

The trained baseline classifier was evaluated using Accuracy, F1-score, a Confusion Matrix, and the Receiver Operating Characteristic curve with its Area Under the Curve.

The poisonous class was treated as the positive class because failing to identify a poisonous mushroom represents the more critical type of classification error.

6.1 Evaluation Metrics

Accuracy measures the proportion of all testing samples that were classified correctly. It provides a useful overall measure because the two target classes are approximately balanced.

Precision for the poisonous class measures the proportion of mushrooms predicted as poisonous that were actually poisonous. High precision indicates that the model rarely incorrectly labels an edible mushroom as poisonous.

Recall for the poisonous class measures the proportion of truly poisonous mushrooms that were identified correctly. Recall is especially important in this project because a poisonous mushroom incorrectly classified as edible represents a potentially dangerous error.

F1-score is the harmonic mean of precision and recall. It summarizes the model's ability to identify poisonous mushrooms while limiting incorrect positive predictions.

The **Confusion Matrix** provides the exact numbers of correct and incorrect predictions for both classes. It makes the different error types visible rather than summarizing performance in a single value.

The **ROC curve** shows the relationship between the True Positive Rate and False Positive Rate at different probability thresholds. The **Area Under the Curve** measures how effectively the model separates edible and poisonous mushrooms across these thresholds. A value close to 1 indicates excellent class separation, while a value close to 0.5 represents performance similar to random guessing.

6.2 Baseline Results

The Logistic Regression baseline produced the following testing results:

Evaluation metric	Result
Mean cross-validation accuracy	99.49%
Test accuracy	99.57%
Poisonous-class precision	100.00%
Poisonous-class recall	99.11%
Poisonous-class F1-score	99.55%
ROC-AUC	99.996%
Correct predictions	1,618
Incorrect predictions	7

The test accuracy was approximately 99.57%, meaning that the classifier correctly predicted 1,618 of the 1,625 testing samples.

The test accuracy is close to the mean cross-validation accuracy of 99.49%. This agreement indicates that the cross-validation result provided a reliable estimate of the model's performance on unseen testing data.

The poisonous-class precision was 100%, meaning that every mushroom predicted as poisonous was actually poisonous. The poisonous-class recall was approximately 99.11%, meaning that the model identified 776 of the 783 poisonous mushrooms in the testing set.

The resulting poisonous-class F1-score was approximately 99.55%, demonstrating a strong balance between precision and recall.

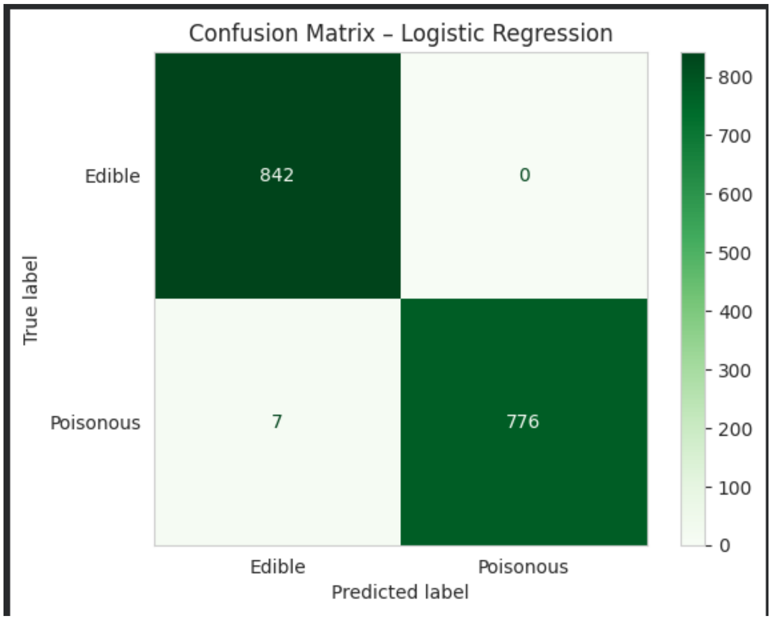

Figure 5: Confusion Matrix for the Logistic Regression baseline classifier.

The Confusion Matrix produced the following results:

True class	Predicted edible	Predicted poisonous
Edible	842	0
Poisonous	7	776

All 842 edible mushrooms were classified correctly. No edible mushrooms were incorrectly classified as poisonous.

Of the 783 truly poisonous mushrooms, 776 were correctly classified as poisonous. However, seven poisonous mushrooms were incorrectly classified as edible.

The model therefore produced no false poisonous warnings, but it missed seven poisonous mushrooms. Although this represents less than 1% of the poisonous testing samples, these errors are important because predicting a poisonous mushroom as edible is the most safety-critical failure in this application.

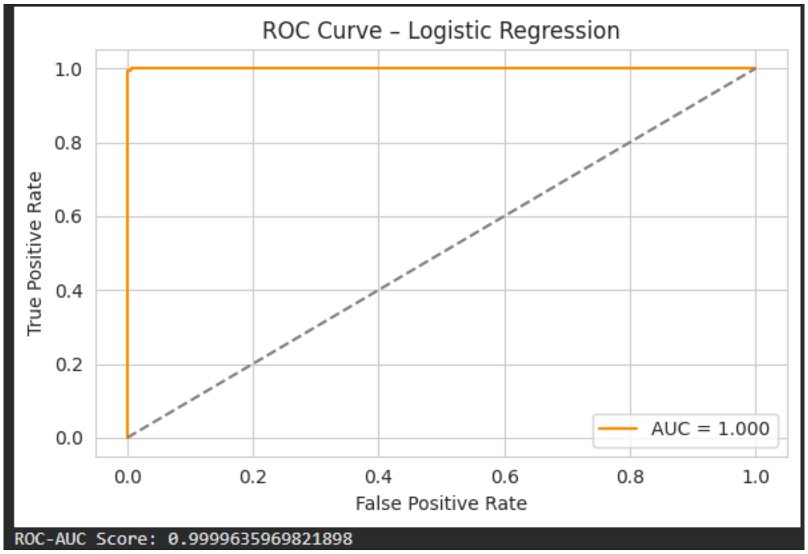

Figure 6: ROC curve for the Logistic Regression baseline classifier.

The ROC-AUC value was approximately 0.99996. The displayed value rounds to 1.000, indicating that the model has an almost perfect ability to rank poisonous mushrooms above edible mushrooms across different probability thresholds.

The ROC curve remains close to the upper-left corner of the graph and lies far above the diagonal random-classification line. This confirms that the model provides excellent separation between the two target classes.

Overall, the results demonstrate that the remaining mushroom features retain strong predictive information even after removing odor and spore-print-color. The high performance is therefore not dependent only on the two excluded obvious features.

7. Preliminary Error Analysis and Limitations

The baseline classifier made seven errors, and all seven had the same form: a poisonous mushroom was predicted as edible. No edible mushroom was predicted as poisonous.

This error direction is important because the cost of the two error types is not equal. Incorrectly classifying an edible mushroom as poisonous would be inconvenient but relatively safe. In contrast, incorrectly classifying a poisonous mushroom as edible could have serious consequences.

The seven false-negative predictions suggest that a default classification threshold may not be ideal for a safety-sensitive application. In future work, the probability threshold could be adjusted to prioritize poisonous-class recall, even if this creates a small number of false poisonous warnings.

Further error analysis should inspect the original characteristics of the seven misclassified mushrooms to identify possible shared patterns. For example, they may contain uncommon feature combinations or characteristics that more closely resemble the edible class.

The preliminary study also has several limitations:

- The results apply to the available UCI dataset and may not generalize to mushroom populations collected from different environments.
- All features are represented as categorical codes, which simplifies real biological variation.
- Some categories contain fewer observations than others.
- Two highly predictive features were removed intentionally, which changes the available information.
- The baseline evaluation includes only Logistic Regression and does not yet compare different model families.
- Very high performance should be interpreted alongside the confusion matrix because overall accuracy can hide safety-critical errors.
- The model should not be used as a real-world mushroom safety tool without external validation and expert biological assessment.

Despite these limitations, the baseline model demonstrates that the preprocessed dataset contains strong and stable predictive patterns. The final project can extend this work by comparing classifiers from different families, tuning model parameters, and analyzing the misclassified observations in greater detail.

8. AI Tool Usage Log

AI tools were used in a limited supporting role for research, clarification, and reviewing the presentation of the project. They were not used to generate the dataset, produce the experimental results, or replace the team's implementation and analysis.

All preprocessing decisions, visualizations, model training, and evaluation were implemented and executed by the team in Google Colab. The reported numerical results were obtained directly from the notebook outputs.

Background Research and Concept Clarification

Purpose:
AI was used to support background research on preprocessing categorical machine learning data. It helped clarify the differences between label encoding and One-Hot Encoding, the purpose of stratified splitting, and the importance of separating training and testing data before learning preprocessing values.

How the information was used:
The research supported the selection of One-Hot Encoding for the categorical features and the use of stratify=y during the train-test split. It also reinforced the decision to calculate the missing-value replacement from the training set only.

Verification:
The concepts were checked against the official Pandas and Scikit-learn documentation before being applied. The implementation was then tested in the notebook, and the final training and testing sets were confirmed to have identical encoded columns and zero missing values.

Research on Suitable EDA Approaches

Purpose:
AI was used to research appropriate visualization approaches for a dataset containing categorical features. It was also used to discuss which feature–target relationships could provide meaningful insights after removing the obvious features.

How the information was used:
Based on this research, count-based bar charts and cross-tabulations were selected. The final EDA examined class distribution, gill size, bruising, and habitat.

Verification:
The team generated all figures from the actual dataset and checked each interpretation against the displayed counts. The meanings of the categorical codes were verified using the official UCI Mushroom Dataset description. The figures were also reviewed to ensure that they satisfied the requirement of exactly three to five meaningful and readable plots.

Research on Evaluation Metrics

Purpose:
AI was used to review the meanings of Accuracy, F1-score, Confusion Matrix, ROC curve, and ROC-AUC in binary classification. Particular attention was given to the importance of treating poisonous mushrooms as the positive class.

How the information was used:
This research helped the team interpret the evaluation results and distinguish between the

two error types. It supported the decision to focus on poisonous-class recall and false-negative errors rather than relying only on overall accuracy.

Verification:
Metric definitions were checked using the official Scikit-learn documentation. All reported values were calculated by the notebook. The team manually confirmed that the Confusion Matrix contained 1,625 testing samples and that the seven model errors were poisonous mushrooms incorrectly predicted as edible.

Critical Reflection

AI tools were useful for quickly reviewing unfamiliar concepts, identifying relevant documentation, and improving the clarity of technical explanations. However, the information provided by AI was treated as preliminary research rather than as an authoritative source.

The team verified important technical decisions using the official UCI, Pandas, and Scikit-learn resources. Dataset inspection, preprocessing, visualization, model implementation, and result interpretation were completed and validated through the project notebook.

Therefore, AI served as a research and clarification assistant, while the project methodology, implementation, experimental outputs, and final conclusions remained the responsibility of the team.

9. Individual Contributions

Team Member	Assigned role	Main contribution
Nadine Radwan	Data and Preprocessing Lead	Data acquisition, inspection, cleaning, feature reduction, splitting, missing-value handling, and encoding
Hisham Mohamed	EDA and Visualization Lead	Descriptive analysis, figures, captions, and interpretations
Ahmed Elsayed	Modeling and Evaluation Lead	Baseline modeling, cross-validation, evaluation, and result analysis

10. Conclusion

This preliminary study established a complete machine learning workflow for classifying mushrooms as edible or poisonous.

The UCI Mushroom Dataset was inspected, cleaned, and prepared for classification. The dataset contained 8,124 samples, no duplicate records, and an approximately balanced target distribution. Missing values were found only in stalk-root and were replaced using the most frequent value obtained from the training data.

To prevent the classification task from becoming overly simple, odor and spore-print-color were removed. The remaining categorical features were converted using One-Hot Encoding, producing 98 aligned model-ready features with no remaining missing values.

The exploratory analysis showed clear relationships between edibility and several mushroom characteristics. Narrow gills, the absence of bruises, and path habitats were more frequently associated with poisonous mushrooms. Broad gills, the presence of bruises, and meadow habitats were more commonly associated with edible mushrooms. However, none of these features independently provided perfect class separation, confirming the importance of combining multiple characteristics.

Logistic Regression was implemented as the single baseline classifier for the preliminary deliverable. It achieved a mean five-fold cross-validation accuracy of 99.49% and a test accuracy of 99.57%. The model also achieved a poisonous-class F1-score of 99.55% and a ROC-AUC of approximately 99.996%.

Out of 1,625 testing samples, the model correctly classified 1,618. All edible mushrooms were classified correctly, while seven poisonous mushrooms were incorrectly predicted as edible. These seven errors are particularly important because false edible predictions represent the highest-risk type of mistake in this problem.

Overall, the results show that the remaining features contain strong predictive information even after removing two highly obvious predictors. The preprocessing strategy was effective, the selected EDA features provided meaningful insights, and Logistic Regression established a strong baseline for the next project stage.

The final stage will extend this work by implementing classifiers from different model families, tuning their parameters, comparing their performance using consistent evaluation metrics, and examining the misclassified samples to determine whether the number of poisonous mushrooms predicted as edible can be reduced.

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